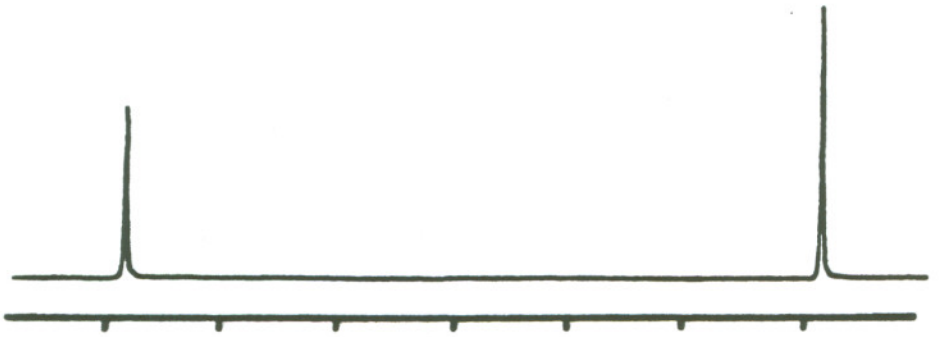
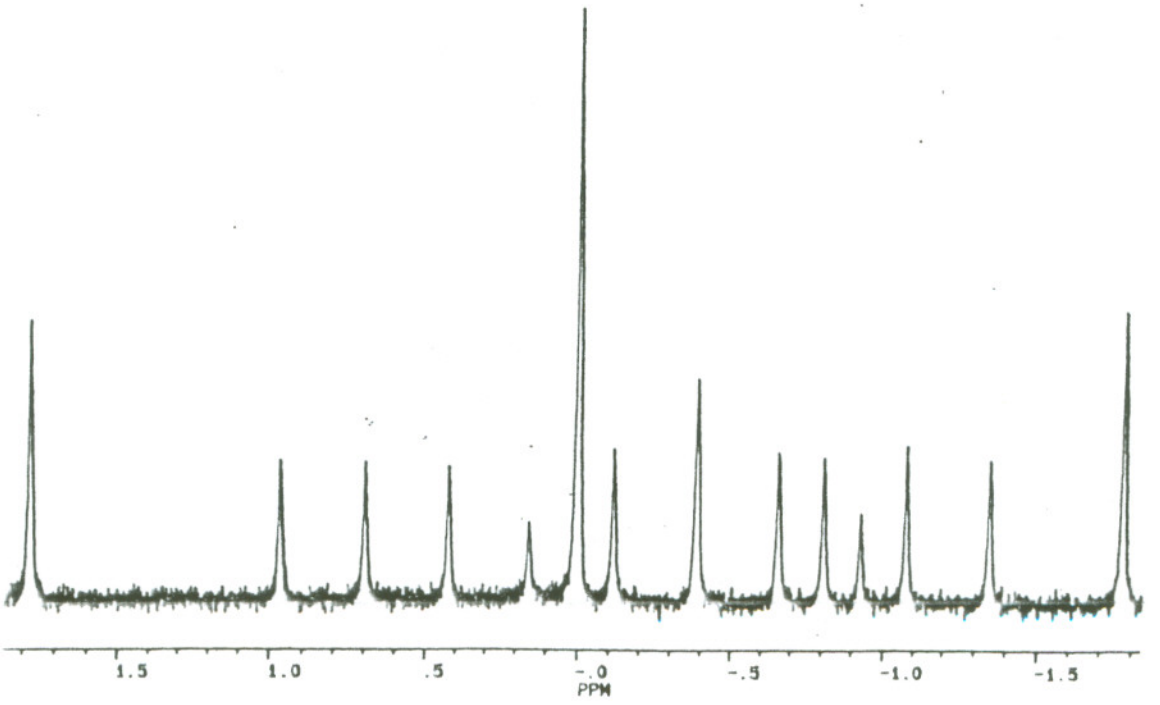
Physical Chemistry Cume

1. Explain the appearance of the titanium spectrum of a solution of $TiCl_4$ obtained at 25.55 MHz in a 9.4 T magnet. The scale markers are 1000 Hz each.



2. The $^{13}C$ spectrum of a mixture of isotopomeric dichloromethanes ($CH_2Cl_2$, $CHDCl_2$ and $CD_2Cl_2$) was obtained without $^{1}H$ broadband decoupling, as shown below. Assign all of the peaks in the spectrum. Discuss the effect of isotope substitution on the $^{13}C$ chemical shifts and coupling constants.



# NMR TABLE
# BY ATOMIC MASS

| NUCLEUS | SPIN | ABUND. | Ho = 7.05T | Ho = 11.75T |
|---|---|---|---|---|
| H-1 | 1/2 | 99.985 | 300.1 | 500.1 |
| H-2 | 1 | $1.5 \times 10^{-2}$ | 46.066 | 76.767 |
| H-3 | 1/2 | — | 320.097 | 533.424 |
| Li-6 | 1 | 7.42 | 44.162 | 73.593 |
| Li-7 | 3/2 | 92.58 | 116.626 | 194.35 |
| Be-9 | -3/2 | 100 | 42.175 | 70.281 |
| B-10 | 3 | 19.58 | 32.25 | 53.741 |
| B-11 | 3/2 | 80.42 | 96.284 | 160.451 |
| C-13 | 1/2 | 1.108 | 75.466 | 125.759 |
| N-14 | 1 | 99.63 | 21.679 | 36.126 |
| N-15 | -1/2 | .37 | 30.409 | 50.674 |
| O-17 | -5/2 | $3.7 \times 10^{-2}$ | 40.684 | 67.797 |
| F-19 | 1/2 | 100 | 282.325 | 470.478 |
| Na-23 | 3/2 | 100 | 79.381 | 132.283 |
| Mg-25 | -5/2 | 10.13 | 18.364 | 30.601 |
| Al-27 | 5/2 | 100 | 78.198 | 130.31 |
| Si-29 | -1/2 | 4.7 | 59.616 | 99.345 |
| P-31 | 1/2 | 100 | 121.482 | 202.442 |
| S-33 | 3/2 | .76 | 23.016 | 38.354 |
| Cl-35 | 3/2 | 75.53 | 29.405 | 49 |
| Cl-37 | 3/2 | 24.47 | 24.472 | 40.78 |
| K-39 | 3/2 | 93.1 | 14.004 | 23.335 |
| K-41 | 3/2 | 6.88 | 7.686 | 12.808 |
| Ca-43 | -7/2 | .145 | 20.192 | 33.647 |
| Sc-45 | 7/2 | 100 | 72.904 | 121.488 |
| Ti-47 | -5/2 | 7.28 | 16.916 | 28.189 |
| Ti-49 | -7/2 | 5.51 | 16.92 | 28.195 |
| V-50 | 6 | .24 | 29.921 | 49.86 |
| V-51 | 7/2 | 99.76 | 78.874 | 131.437 |
| Cr-53 | -3/2 | 9.55 | 16.962 | 28.266 |
| Mn-55 | 5/2 | 100 | 74.017 | 123.344 |
| Fe-57 | 1/2 | 2.19 | 9.697 | 16.158 |
| Co-59 | 7/2 | 100 | 70.867 | 118.093 |
| Ni-61 | -3/2 | 1.19 | 26.818 | 44.689 |
| Cu-63 | 3/2 | 69.09 | 79.544 | 132.553 |
| Cu-65 | 3/2 | 30.91 | 85.211 | 141.997 |
| Zn-67 | 5/2 | 4.11 | 18.771 | 31.278 |
| Ga-69 | 3/2 | 60.4 | 72.037 | 120.045 |
| Ga-71 | 3/2 | 39.6 | 91.519 | 152.51 |
| Ge-73 | -9/2 | 7.76 | 10.468 | 17.444 |
| As-75 | 3/2 | 100 | 51.398 | 85.65 |
| Se-77 | 1/2 | 7.58 | 57.221 | 95.353 |
| Br-79 | 3/2 | 50.54 | 75.187 | 125.294 |
| Br-81 | 3/2 | 49.46 | 81.045 | 135.045 |
| Rb-85 | 5/2 | 72.15 | 28.975 | 48.285 |
| Rb-87 | 3/2 | 27.85 | 98.194 | 163.634 |
| Sr-87 | -9/2 | 7.02 | 13.006 | 21.673 |
| Y-89 | -1/2 | 100 | 14.703 | 24.5 |
| Zr-91 | -5/2 | 1.23 | 28.001 | 46.66 |
| Nb-93 | 9/2 | 100 | 73.355 | 122.241 |
| Mo-95 | 5/2 | 15.72 | 19.553 | 32.583 |
| Mo-97 | -5/2 | 9.46 | 19.962 | 33.264 |
| Ru-99 | -3/2 | 12.72 | 13.852 | 23.062 |
| Ru-101 | -5/2 | 17.07 | 15.522 | 25.862 |
| Rh-103 | -1/2 | 100 | 9.445 | 15.739 |
| Pd-105 | -5/2 | 22.23 | 13.745 | 22.903 |
| Ag-107 | -1/2 | 51.82 | 12.144 | 20.237 |
| Ag-109 | -1/2 | 48.18 | 13.961 | 23.276 |
| Cd-111 | -1/2 | 12.75 | 63.635 | 106.043 |
| Cd-113 | -1/2 | 12.26 | 66.574 | 110.941 |
| In-113 | 9/2 | 4.28 | 65.622 | 109.354 |
| In-115 | 9/2 | 95.72 | 65.764 | 109.592 |
| Sn-117 | -1/2 | 7.61 | 106.913 | 178.164 |
| Sn-119 | -1/2 | 58 | 111.854 | 186.398 |
| Sb-121 | 5/2 | 7.25 | 71.818 | 119.68 |
| Sb-123 | 7/2 | 42.75 | 78.663 | 131.084 |
| Te-123 | -1/2 | .87 | 38.891 | 64.809 |
| Te-125 | -1/2 | 6.99 | 94.804 | 157.984 |
| I-127 | 5/2 | 100 | 60.042 | 100.056 |
| Xe-129 | 1/2 | 26.44 | 83.011 | 138.332 |
| Cs-133 | 7/2 | 100 | 39.364 | 65.597 |
| Ba-135 | 3/2 | 6.59 | 29.813 | 49.68 |
| Ba-137 | 3/2 | 11.32 | 33.351 | 55.576 |
| La-139 | 7/2 | 99.911 | 42.393 | 70.646 |
| Pr-141 | 5/2 | 100 | 88.107 | 146.826 |
| Nd-143 | -7/2 | 12.17 | 16.317 | 27.192 |
| Nd-145 | -7/2 | 8.3 | 10.009 | 16.679 |
| Sm-147 | -7/2 | 14.97 | 12.406 | 20.673 |
| Sm-149 | -7/2 | 13.83 | 9.868 | 16.443 |
| Eu-151 | 5/2 | 47.82 | 74.426 | 124.026 |
| Eu-153 | 5/2 | 52.18 | 32.865 | 54.767 |
| Gd-155 | -3/2 | 14.73 | 11.278 | 18.794 |
| Gd-157 | -3/2 | 15.68 | 14.097 | 23.492 |
| Tb-159 | 3/2 | 100 | 68.089 | 113.467 |
| Dy-161 | -5/2 | 18.88 | 9.868 | 16.443 |
| Dy-163 | 5/2 | 24.97 | 14.097 | 23.492 |
| Ho-165 | 7/2 | 100 | 61.534 | 102.542 |
| Er-167 | -7/2 | 22.94 | 8.67 | 14.447 |
| Tm-169 | -1/2 | 100 | 24.81 | 41.345 |
| Yb-171 | 1/2 | 14.31 | 52.857 | 88.083 |
| Yb-173 | -5/2 | 16.13 | 14.562 | 24.265 |
| Lu-175 | 7/2 | 97.41 | 34.256 | 57.085 |
| Hf-177 | 7/2 | 18.5 | 9.163 | 15.269 |
| Hf-179 | -9/2 | 13.75 | 5.639 | 9.397 |
| Ta-181 | 7/2 | 99.988 | 35.92 | 58.858 |
| W-183 | 1/2 | 14.4 | 12.487 | 20.809 |
| Re-185 | 5/2 | 37.07 | 67.564 | 112.591 |
| Re-187 | 5/2 | 62.93 | 68.256 | 113.744 |
| Os-187 | 1/2 | 1.64 | 6.912 | 11.517 |
| Os-189 | 3/2 | 16.1 | 23.284 | 38.801 |
| Ir-191 | 3/2 | 37.3 | 5.158 | 8.596 |
| Ir-193 | 3/2 | 62.7 | 5.616 | 9.359 |
| Pt-195 | 1/2 | 33.8 | 64.516 | 107.511 |
| Au-197 | 3/2 | 100 | 5.14 | 8.564 |
| Hg-199 | 1/2 | 16.84 | 53.499 | 89.153 |
| Hg-201 | -3/2 | 13.22 | 19.806 | 33.005 |
| Tl-203 | 1/2 | 29.5 | 171.506 | 285.805 |
| Tl-205 | 1/2 | 70.5 | 173.108 | 288.474 |
| Pb-207 | 1/2 | 22.6 | 62.786 | 104.629 |
| Bi-209 | 9/2 | 100 | 48.225 | 80.364 |

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